CoRoT-2b

R-0208

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_240724` · created 2026-07-16T23:22:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460515.8148 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.048
Transit depth (Rp/Rs)^2	0.00086 +/- 0.028 [%]
Orbital Inclination (inc)	84.69 +/- 2.02
Airmass coefficient 1 (a1)	0.82 +/- 0.058
Airmass coefficient 2 (a2)	0.17 +/- 0.12
Scatter in the residuals of the lightcurve fit is	7.22 %
Best Comparison Star	#3 - [97, 115]
Optimal Aperture	2.83
Optimal Annulus	10.86
Transit Duration (day)	0.065 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.003 ± 0.048 vs 0.1667 (deviation 3.02σ)
Duration fit vs published	0.065 d vs 0.095 d (deviation 1.4σ)
Mid-time O–C	4.5 min
Depth SNR	0.1
Residual scatter	7.22 %

Light curve

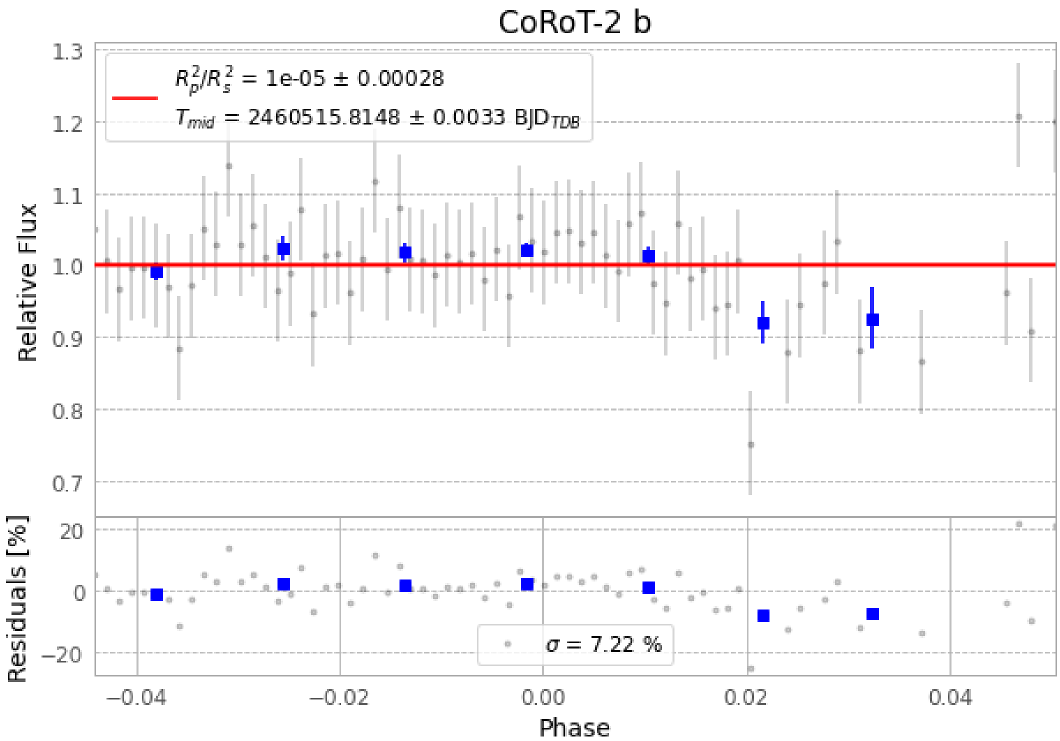

Figure 1 — FinalLightCurve_CoRoT-2 b_2024-07-23.png (SHA-256 de8279b97dbc7673…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [291, 195], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ffcffefe69e199ef…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e`

Data provenance

80 FITS frames (53 MB), `CoRoT-2240724053317.FITS` → `CoRoT-2240724093015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240724.log (e3f6a299d568…), FinalLightCurve_CoRoT-2 b_2024-07-23.png (de8279b97dbc…), FinalLightCurve_CoRoT-2 b_2024-07-23.csv (952e78ad7bc2…)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 23:22Z.